

# Gulf of Thailand model validation

| Field                              | Entry                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Region                             | Gulf of Thailand, LME_035; source 10–50 m shelf applied across the whole LME. <a href="#">Regional workbook</a>                                                                                                                                                                                                                                                                                                                                                                             |
| Catch source                       | Sea Around Us v50.1; local 1950–2019 catch; reference: 2019 landings. <a href="#">Original catch</a>                                                                                                                                                                                                                                                                                                                                                                                        |
| Selected article                   | Christensen, V. (1998). Fishery-induced changes in a marine ecosystem: insight from models of the Gulf of Thailand. Journal of Fish Biology 53 (Suppl. A): 128–142. Author full HTML read; original PDF unavailable. <a href="#">DOI</a>   <a href="#">Author full text</a>   <a href="#">Retained source facts</a>                                                                                                                                                                         |
| Other known articles               | Pauly & Christensen (1993), Stratified models of large marine ecosystems: application to the South China Sea — acknowledged 10–50 m Gulf predecessor. Vibunpant et al. (2003), Trophic model of the coastal fisheries ecosystem in the Gulf of Thailand — separate 40-group 1973 model. Premcharoen (2012), Mae Klong estuary — subregional alternative. <a href="#">1993 predecessor</a>   <a href="#">1973 source</a>   <a href="#">Estuary source</a>                                    |
| Selected model                     | 35_412_Gulf_of_Thailand_(1963); scientific payload 1980 despite inherited 1963 filename. EcoBase 412 description and all 11 Table II biomass comparisons identify 1980. 29 source ecological groups (+1 computational import); 10–50 m shelf. <a href="#">Canonical model</a>   <a href="#">Identity evidence</a>                                                                                                                                                                           |
| Other models from the same article | Article compares 1963 and 1980 baselines. Selected canonical file contains 1980; a separate verified complete 1963 parameter/diet set is not supplied. 1993 predecessor and 1973 candidate have different group structures.                                                                                                                                                                                                                                                                 |
| Selection rationale                | Article — ?<br>Model — Retained selected EcoBase 412 payload; independent identity evidence establishes 1980. Whole-LME/year application remains provisional; scientific approval is not established by finite coefficients.                                                                                                                                                                                                                                                                |
| Model extraction                   | Available source identity, habitat, illustrative membership and 11 biomasses were checked. Complete published B/PB/QB/EE/diet inputs are unavailable. Accepted canonical export, BA, diet, routing and runtime settings are retained; exact loaded groups agree. Zero juvenile exports are canonical values, not observed regional stage catches. <a href="#">Source limitations</a>   <a href="#">Loaded proof</a><br>Retained departures and evidence limits: <a href="#">model notes</a> |
| GE diagnostics                     | WARN: Jellyfish (18) and M. mammals (1) have EE = 0.<br>No negative SPPR entries across the basal-source columns.<br>rho_living = 0.2400<br>b = 0.2712<br>Detritus (29): SPPR = 2.5802                                                                                                                                                                                                                                                                                                      |
| TE diagnostics                     | WARN: Jellyfish (18) and M. mammals (1) have EE = 0.<br>No negative SPPR entries across the basal-source columns.<br>rho_living = 0.6142<br>Detritus (29): SPPR = 0.6600                                                                                                                                                                                                                                                                                                                    |

| Field                              | Entry                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SPPR calculation</b>            | <p>Removed groups from calculations:</p> <ul style="list-style-type: none"> <li>- M. mammals (GE=0.10%, SPPR_GE=110K)</li> </ul>                                                                                                                                                                                                                                                             |
| <b>Geographic fit</b>              | <p>A. Region covered by the focal 10–50 m study: not determined.</p> <p>B. Focal study area covered by the region: not determined.</p> <p>The original focal boundary is unavailable; the contextual estimates are shown with the figures below.</p>                                                                                                                                         |
| <b>Temporal fit</b>                | <p>Static 1980 coefficients and fixed composition/stage proxies apply across 1950–2019 whole-LME catches; reference 2019 is 39 years later. Source shelf and reporting-population differences are not reconstructed annual ecosystem dynamics.</p>                                                                                                                                           |
| <b>Other</b>                       | <p>GE, TE and With Egestion remain WARN. TE repairs consumed-production routing for two EE = 0 groups (marine mammals, jellyfish) under accepted runtime settings. All 270 direct source entries are finite/nonnegative, including unfished groups; six symbolic methods remain failed. Source fidelity and spatial/temporal application limits remain. <a href="#">Full diagnostics</a></p> |
| <b>Open issues and next action</b> | <p>[Researcher: open issues and next actions]</p>                                                                                                                                                                                                                                                                                                                                            |
| <b>Review and reproducibility</b>  | <p>Researcher name: Ido Carmel   review date: 08/10/2026</p> <p><b>MODEL VALIDATED</b></p>                                                                                                                                                                                                                                                                                                   |

## Taxon mapping and coverage

Reference: 2019 landings; all taxon labels, including zero catch and unidentified categories; no group filter.

29 source ecological groups (+1 computational import); 247 catch taxa.

Simple-chain PPR totals 192,544,832.38 t C in 2019.

33 labels lack a classic TL/coefficient; all have zero 2019 landings and contribute zero annual PPR.

Method: independent simple trophic chain.

[All taxon mappings](#) and [descriptive Sources sheet](#)

| Overall confidence | Taxa (n) | Catch (%) | Simple-chain PPR (%) |
|--------------------|----------|-----------|----------------------|
| High               | 99       | 51%       | 33%                  |
| Medium             | 103      | 27%       | 45%                  |
| Low                | 11       | 8.7%      | 11%                  |
| Very low           | 34       | 13%       | 12%                  |
| Unresolved         | 0        | 0%        | 0%                   |

Overall confidence uses the weaker required membership or allocation component.

### Group assignment rules

| Plain-language rule                                           | Confidence | PPR percentage |
|---------------------------------------------------------------|------------|----------------|
| Supported coastal/guild or historical ecological extension.   | Medium     | 26%            |
| Unambiguous documented generic taxonomic group fit.           | High       | 20%            |
| Explicit focal source taxon or genus assignment.              | High       | 19%            |
| Assumed complete eligible family or suborder set.             | Medium     | 16%            |
| Broad reporting category approximated over compatible pools.  | Very low   | 12%            |
| Related-member extension from source representatives.         | Medium     | 6.0%           |
| Verified source spelling or historical-name bridge.           | High       | 0.27%          |
| Closest represented analogue with material source mismatches. | Very low   | 0.27%          |
| Partial or competing ecological/taxonomic fit.                | Low        | 0.056%         |

Membership evidence: [Christensen \(1998\), p.130 and model identity](#), [Pauly & Christensen \(1993\), Table 2](#) and [all taxon decisions](#)

## Allocation weight rules

| Plain-language rule                                                    | Confidence | PPR percentage |
|------------------------------------------------------------------------|------------|----------------|
| One reviewed group receives 100%; no split.                            | High       | 66%            |
| Complete 1980 canonical export proportions, including zero candidates. | Medium     | 23%            |
| Retained historical identified-catch proxy with population mismatch.   | Low        | 10%            |

Each percentage is the independent simple-chain PPR associated with taxa using that rule. The 1980 model-catch proportions, including zero candidates, are fixed composition proxies. Historical identified-catch proxies remain Low because donor populations and gear differ. [Weights and donor evidence](#)

Unresolved taxa: none. The 34 Very low decisions cover 12.9493% of landings and 12.0040% of simple-chain PPR. Their specific uncertainties appear below. Very low decisions contribute 14.12% of GE, 12.01% of TE and 13.86% of With Egestion all-source PPR; Carangidae alone contributes 7.93% of GE. [Method-specific exposure](#)

## Very low decisions

| Affected taxa                                                                                                   | Why confidence is very low                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Carangidae                                                                                                      | Scad explicitly includes Decapterus; larger predators use the large-piscivore stage pair as an analogue. The reported jacks/pompanos mixture is unknown. Model-catch proportions assume fixed 1980 composition.                                                 |
| Marine fishes not identified                                                                                    | Unidentified bony fish may span named, residual and juvenile pools. Separate reported categories do not establish exclusions; species and stage composition is unknown. Model-catch proportions are a fixed 1980 proxy.                                         |
| Serranidae                                                                                                      | The historical basses/groupers category includes Epinephelus, now Epinephelidae, while other basses are unlisted. The grouper/snapper stage union is a proxy for an unresolved taxonomic and feeding mixture; model-catch proportions are assumed.              |
| Alepes                                                                                                          | Scad provides a coastal schooling Decapterus connection, but reef habitat, feeding and species composition differ. No direct source membership is documented; the whole taxon is placed in Scad.                                                                |
| Marine groundfishes not identified                                                                              | Plausible demersal and juvenile pools are retained, but the large-piscivore pool also contains pelagic Chirocentrus/Sphyræna. Unknown residual composition and mixed guilds require fixed 1980 model-catch proportions.                                         |
| Caesio; Caesionidae                                                                                             | Small pelagics provides a coastal schooling sardine/anchovy connection. Reef habitat, feeding and species composition differ; fusiliers are not directly named. The whole taxon is placed in that analogue.                                                     |
| Acanthurus; Mugil; Mugilidae; Scarus; Scatophagus; Scatophagus argus; Siganidae; Siganus; Siganus canaliculatus | These unlisted coastal herbivores/omnivores have a residual-catch connection to O.fish (trash fish). Its species composition and feeding fit are unverified. Named piscivore/benthivore pools offer weaker connections; each whole taxon is assigned to O.fish. |
| Perciformes                                                                                                     | The historical perch-like reporting category spans scombrids, carangids and reef/demersal guilds; the current narrow order does not define that mixture. Source piscivore pools also include other orders. Model-catch proportions assume fixed composition.    |

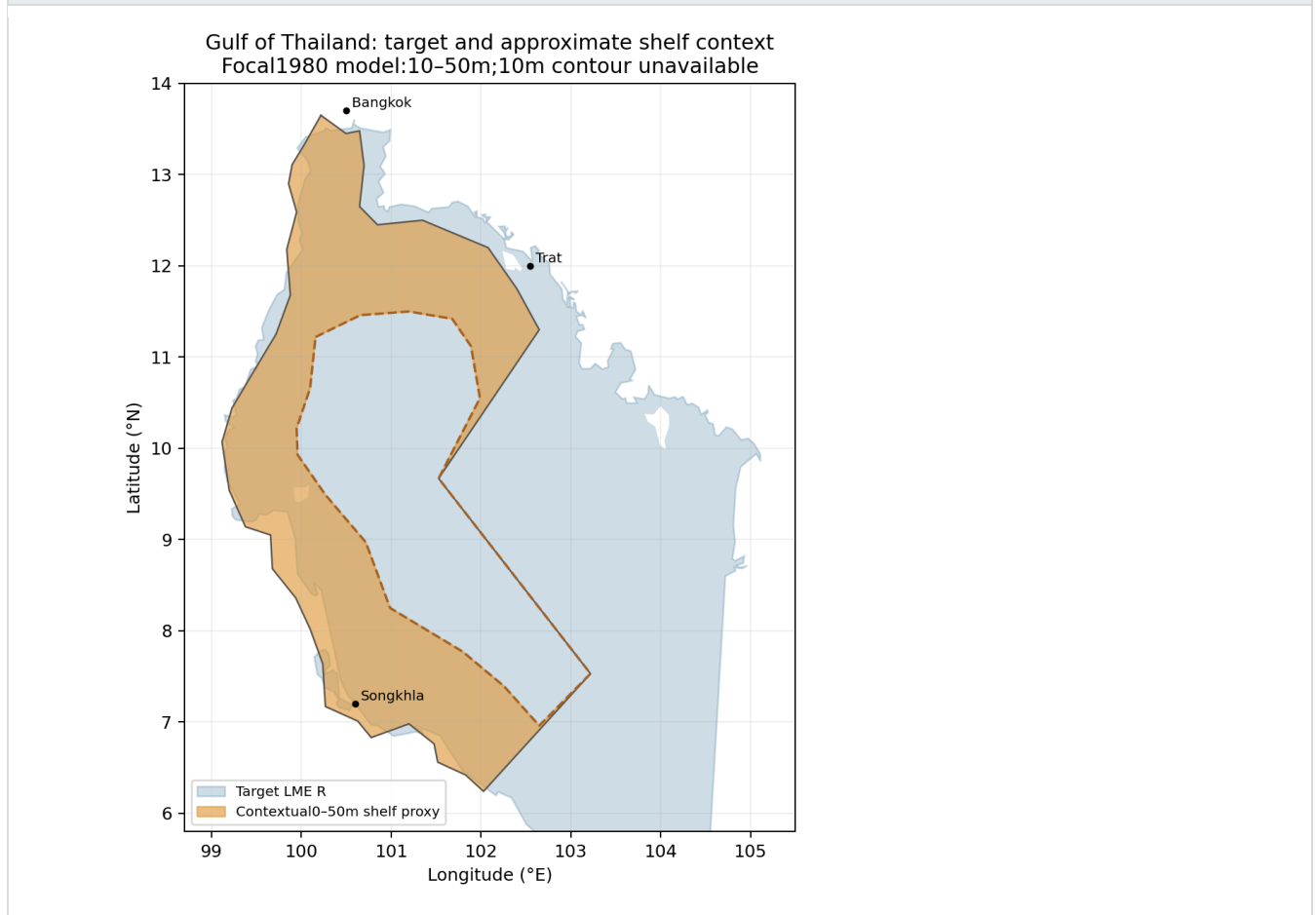
| Affected taxa                                                                                                                                              | Why confidence is very low                                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chondrichthyes                                                                                                                                             | Shark stages and rays approximate the reporting category; chimaeras have no documented source group. Species/stage composition is unknown. Historical whole-guild donor fractions have country, gear and period mismatches and remain Low allocation proxies.                                |
| Decapoda; Malacostraca; Miscellaneous aquatic invertebrates; Miscellaneous marine crustaceans; Mollusca                                                    | Compatible named invertebrate, benthic and planktonic stage pools provide a practical connection. The full reported composition is unenumerated, so eligible pools and species/stage mix are approximate. Fixed 1980 model-catch proportions are assumed.                                    |
| Coryphaena hippurus                                                                                                                                        | The large-predator stage pair connects to source coastal piscivores, but dolphinfish is unlisted and Tuna is a plausible competing pelagic analogue. Surface habitat and source composition differ. The accepted stage split uses fixed 1980 model-catch proportions.                        |
| Istiompax indica; Istiophoridae; Istiophorus; Istiophorus platypterus; Kajikia audax; Makaira; Makaira mazara; Tetrapturus angustirostris; Xiphias gladius | The 1993 predecessor places billfish with tuna in Large pelagics, supporting a crosswalk to focal Tuna. Focal members are coastal tuna examples; oceanic/deeper habitat and narrower taxonomy limit the analogy. No focal billfish membership or donor coefficients are implied.             |
| Marine pelagic fishes not identified                                                                                                                       | Named pelagic pools and mixed guilds containing Chirocentrus/Sphyræna or Pampus provide positive source connections. The full residual species/stage mixture is unknown; explicit demersal exclusions do not remove mixed-guild uncertainty. Fixed 1980 model-catch proportions are assumed. |

# Geographic evidence screenshots

Target region R is the Sea Around Us LME. The focal source specifies a 10–50 m shelf domain, but its original study boundary is unavailable. Pauly & Chuenpagdee (2003), Figure 14-1, provides contextual geography; its illustrative trace does not establish the focal study extent.

## Region area

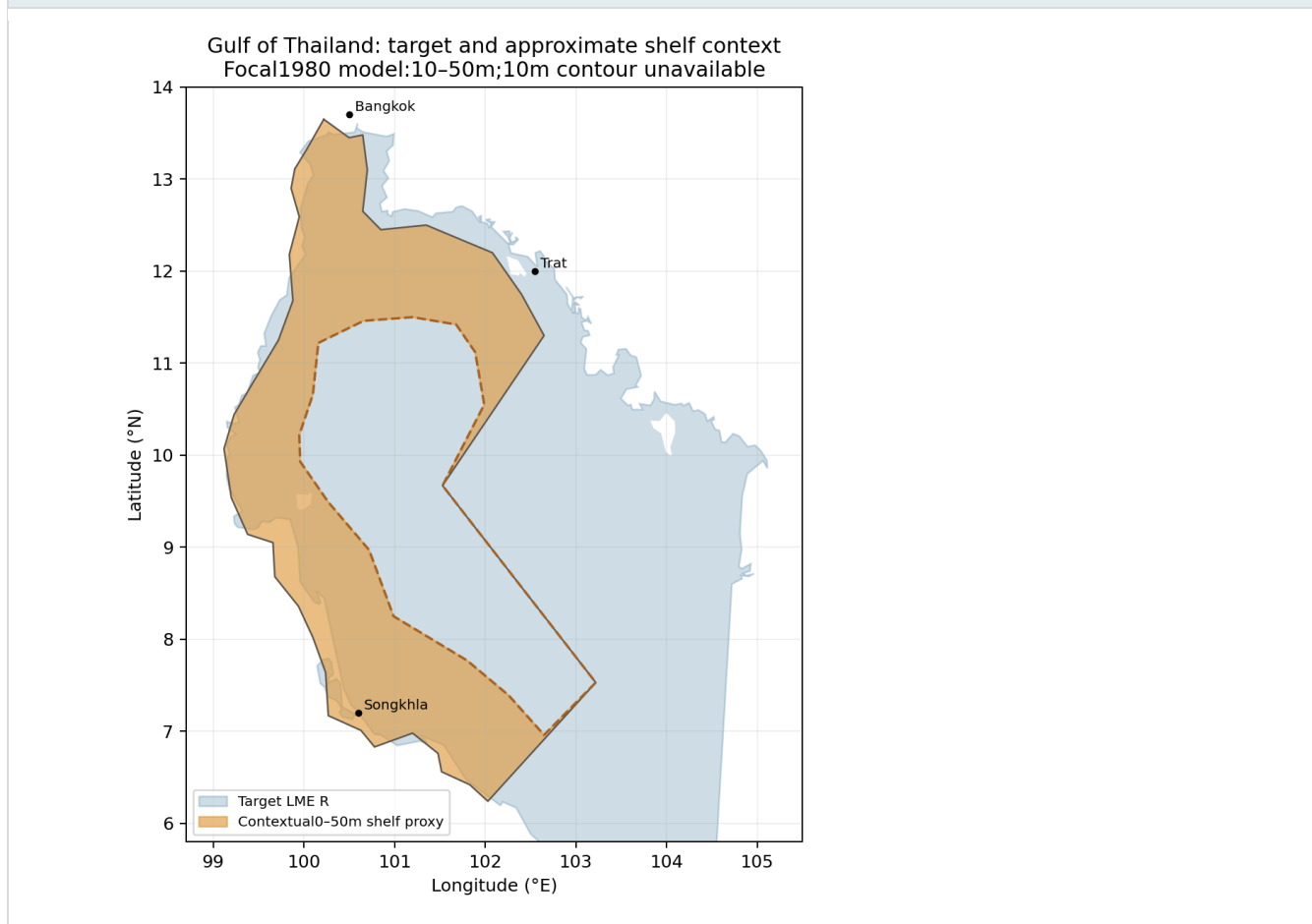
### Gulf of Thailand target and shelf context



Target R: 387,239 km<sup>2</sup>. Blue shows the target; orange shows an approximate contextual shelf trace within the lineage map boundary. Coastline and islands are simplified. [Target polygon](#) | [Context trace](#)

## Study area in the article

### Illustrative depth context from lineage map



Contextual trace: 112,695 km<sup>2</sup>; intersection: 102,520 km<sup>2</sup>. Contextual A  $\approx$  26.5%; B  $\approx$  91.0%. The reported larger 0–50 m band ( $\approx$ 150,000 km<sup>2</sup>,  $\approx$ 38.7% of R) is separate area context. Neither identifies the missing focal 10 m boundary or full study extent. [Lineage Figure14-1](#) | [Calibration and limits](#)

Contextual Figure 14-1 from Pauly & Chuenpagdee (2003), printed p.338 / PDF p.2. Its 50 m contour and angular location boundary do not constitute the original focal study map.

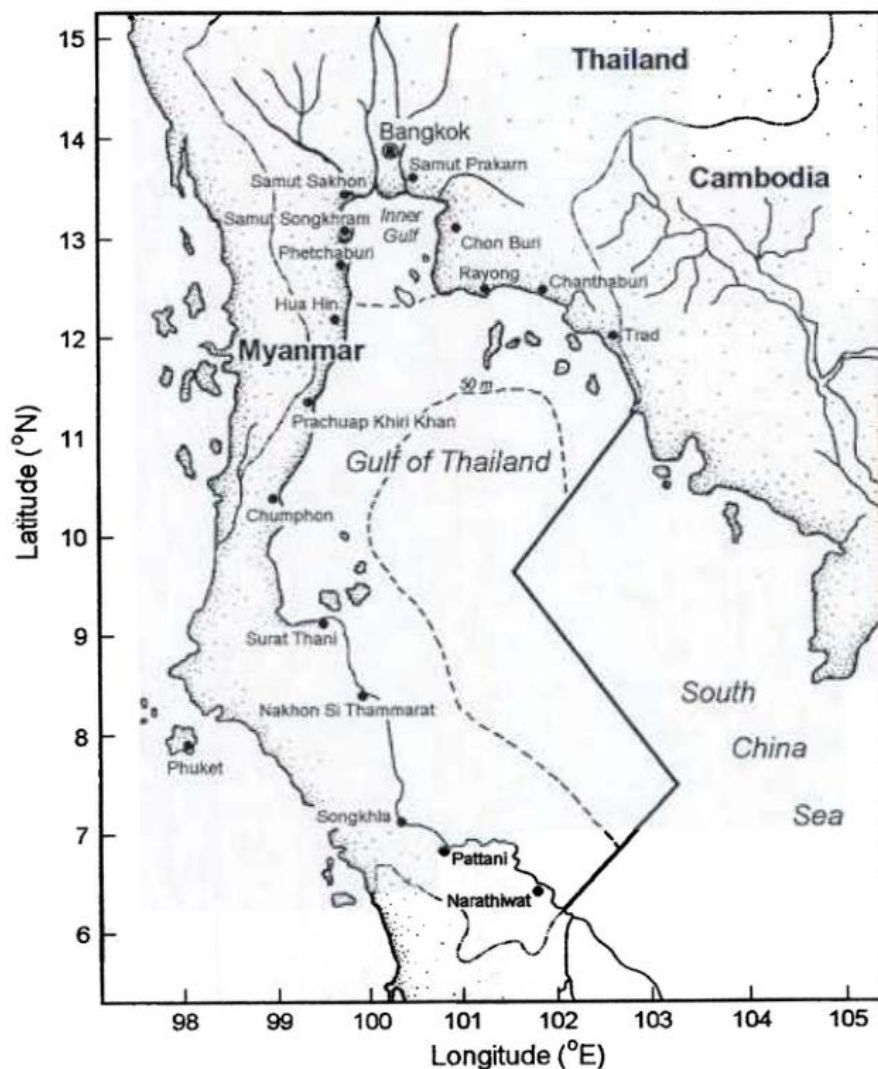


Figure 14-1. Map of the Gulf of Thailand, highlighting features mentioned in the text.